DATA ANALYTICS PROJECT REPORT

Online Retail
Sales Analytics

Exploratory Analysis of Transaction-Level Retail Sales Data

Dataset period analysed: 1 December 2010 – 9 December 2011
Records analysed (post-cleaning): 401,604 transactions
Tools: Python · Pandas · Matplotlib · Jupyter Notebook

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SECTION 02

Executive Summary

This report presents an end-to-end exploratory analysis of a transactional online retail dataset covering the period from 1 December 2010 to 9 December 2011. The raw dataset contained 541,909 records across 8 columns. Following a structured data cleaning process (removal of cancelled/returned transactions, invalid records, duplicates, and rows with missing customer or product identifiers), the analysis was carried out on a cleaned dataset of 401,604 records.

The cleaned data was used to compute a derived Revenue metric (Quantity × UnitPrice), resulting in total recorded revenue of **£8,278,519.42** across the observed period. Monthly revenue and quantity trends were examined, and the top 10 revenue-generating products were identified. The findings below are based strictly on the computations performed in the accompanying notebooks; no additional statistics or claims beyond these computations are included.

SECTION 03

Business Objective

The objective of this project was to analyse transaction-level online retail sales data in order to understand overall sales performance, identify monthly revenue and quantity trends across the observed period, and determine which products contributed the most recorded revenue. The analysis is descriptive and exploratory in nature and is intended to surface patterns present in the data rather than to establish causal business drivers.

SECTION 04

Dataset Overview

The source dataset is a transactional online retail dataset consisting of 541,909 rows and 8 columns, spanning invoice-level purchase records from a UK-based online retailer. The table below describes each column as it appears in the raw data.

Column	Description
InvoiceNo	Unique identifier for each transaction/invoice
StockCode	Unique identifier for each product
Description	Product name/description
Quantity	Number of units purchased per transaction line
InvoiceDate	Date and time the transaction occurred
UnitPrice	Price per unit, in GBP (£)
CustomerID	Unique identifier for each customer
Country	Country associated with the customer

SECTION 05

Data Understanding

An initial inspection of the raw dataset (documented in **01_data_understanding.ipynb**) was performed to assess its structure, completeness, and value ranges before any cleaning was applied.

Missing Values (Raw Dataset)

Column	Missing Values
InvoiceNo	25,900
StockCode	4,070
Description	4,223
Quantity	722
InvoiceDate	23,260
UnitPrice	1,630
CustomerID	4,372
Country	38

Descriptive Statistics (Raw Dataset)

Field	Mean	Median	Minimum	Maximum
Quantity	9.552	3	-80,995	80,995
UnitPrice (£)	4.611	2.08	-11,062.06	38,970

Key Observations

- CustomerID had a substantial number of missing values (4,372 rows).
- Description also contained missing values (4,223 rows).
- Quantity contained negative values, ranging down to -80,995.
- UnitPrice contained negative values, with a minimum of -£11,062.06.
- Country was heavily dominated by the United Kingdom.
- The dataset contained approximately 25,900 unique invoice numbers.

SECTION 06

Data Quality Findings

The data understanding phase surfaced several data quality issues that required resolution before the dataset could be used for sales analysis:

- **Missing identifiers:** Both CustomerID and Description had non-trivial volumes of missing values, which would limit the reliability of customer-level and product-level analysis if left unaddressed.
- **Negative Quantity values:** 10,624 records had negative Quantity. Of these, 9,288 had an InvoiceNo beginning with the letter "C", which is strongly associated with order cancellations or returns. The remaining negative-quantity records were also non-standard, frequently missing Description/CustomerID and carrying a UnitPrice of 0.
- **Negative UnitPrice values:** Two records had negative UnitPrice. Both were identified as "Adjust bad debt" accounting adjustment entries and had no associated CustomerID.
- **Duplicate rows:** 5,268 exact duplicate rows were present in the raw dataset.
- **Country concentration:** The Country field was heavily dominated by the United Kingdom, which should be kept in mind when interpreting country-level patterns.

SECTION 07

Data Cleaning Methodology

Data cleaning was performed in **02_data_cleaning.ipynb**. The steps below were applied sequentially to the raw dataset.

1. Identified two records with negative UnitPrice values.
2. Confirmed both were “Adjust bad debt” accounting adjustment records with no CustomerID, and excluded them, since the project focuses on retail sales transactions rather than accounting adjustments.
3. Investigated the 10,624 records with negative Quantity values.
4. Confirmed 9,288 of these had an InvoiceNo beginning with “C”, consistent with cancellations/returns.
5. Noted that the remaining negative-quantity records were also non-standard, often missing Description/CustomerID with UnitPrice = 0.
6. Excluded all records with Quantity ≤ 0, since the analysis is focused on completed sales.
7. Identified and removed 5,268 exact duplicate rows.
8. Removed rows with missing CustomerID, since customer-level analysis requires an identifiable customer.
9. Removed rows with missing Description, since product-level analysis requires product information.
10. Reset the DataFrame index after cleaning.

Result of Cleaning

Metric	Value
Rows (raw)	541,909
Rows (cleaned)	401,604
Columns	8 (unchanged)
Exported file	online_retail_cleaned.xlsx

SECTION 08

Exploratory Data Analysis

Exploratory analysis was carried out in **03_exploratory_analysis.ipynb** on the cleaned dataset. A Revenue field was derived as:

$$\textbf{Revenue} = \textbf{Quantity} \times \textbf{UnitPrice}$$

Total recorded revenue across the cleaned dataset was **£8,278,519.42**, covering the period from 1 December 2010 to 9 December 2011.

Monthly Revenue

Month	Revenue	Month	Revenue
Dec 2010	£552,372.86	Jul 2011	£573,112.32
Jan 2011	£473,731.90	Aug 2011	£615,078.09
Feb 2011	£435,534.07	Sep 2011	£929,356.23
Mar 2011	£578,576.21	Oct 2011	£973,306.38
Apr 2011	£425,222.67	Nov 2011	£1,126,815.07
May 2011	£647,011.67	Dec 2011*	£341,539.43
Jun 2011	£606,862.52		

** December 2011 is incomplete: the dataset ends on 9 December 2011 and should not be directly compared with complete calendar months.*

Monthly Quantity Sold

Month	Quantity	Month	Quantity
Dec 2010	295,177	Jul 2011	361,359
Jan 2011	268,755	Aug 2011	385,865
Feb 2011	262,243	Sep 2011	536,350
Mar 2011	343,095	Oct 2011	568,898
Apr 2011	277,730	Nov 2011	666,813
May 2011	367,115	Dec 2011*	203,212
Jun 2011	356,239		

** December 2011 is incomplete (data ends 9 December 2011).*

Top 10 Revenue-Generating Products

Rank	Product Description	Revenue
1	REGENCY CAKESTAND 3 TIER	£132,567.70
2	WHITE HANGING HEART T-LIGHT HOLDER	£93,767.80
3	JUMBO BAG RED RETROSPOT	£83,056.52
4	PARTY BUNTING	£67,628.43
5	POSTAGE*	£66,710.24
6	ASSORTED COLOUR BIRD ORNAMENT	£56,331.91
7	RABBIT NIGHT LIGHT	£51,042.84
8	CHILLI LIGHTS	£45,915.41
9	PAPER CHAIN KIT 50'S CHRISTMAS	£41,423.78
10	PICNIC BASKET WICKER 60 PIECES	£39,619.50

** POSTAGE is a shipping-related line item rather than a physical product and should be interpreted with this distinction in mind.*

SECTION 09

Key Findings

- November 2011 was the highest full month by revenue, at approximately £1.13M.
- Revenue increased substantially from September through November 2011.
- April 2011 was the lowest complete month by revenue, at approximately £425K.
- December 2011 is incomplete (data ends 9 December 2011) and should not be compared directly against complete months.
- Monthly revenue and monthly quantity sold follow a similar upward pattern from September to November 2011, suggesting the revenue increase was strongly associated with higher sales volume over that period. This is an observed relationship, not evidence of causation.
- REGENCY CAKESTAND 3 TIER generated the highest recorded revenue among all product descriptions (£132,567.70).
- POSTAGE appears within the top 10 revenue-generating line items; as it is not a conventional physical product, it should be treated separately from merchandise performance.

SECTION 10

Business Interpretation

The interpretations below are drawn directly from the patterns observed in the cleaned data and are clearly distinguished from any underlying assumptions.

Observations (directly supported by the data)

- Revenue and quantity sold rose together from September through November 2011, indicating that higher sales volume was the primary driver of the revenue increase in that window.
- November 2011 was the strongest complete month in the observed period, while April 2011 was the weakest.
- A small number of products, led by REGENCY CAKESTAND 3 TIER, account for a disproportionately large share of the top-10 revenue total.

Assumption (not verified within this analysis)

The observed correlation between quantity sold and revenue is not evidence of a causal relationship; other factors could contribute to both trends moving together. Establishing causation, or identifying the specific business or seasonal factors behind the September–November increase, was outside the scope of the analysis performed and would require further investigation.

SECTION 11

Limitations

- December 2011 is incomplete (the dataset ends on 9 December 2011); it is excluded from month-over-month comparisons involving complete calendar months.
- Rows with missing CustomerID were removed, which excludes any transactions that could not be attributed to an identifiable customer from the analysis.
- Rows with missing Description were removed, which excludes transactions lacking product information from the analysis.
- Records with Quantity ≤ 0 (cancellations, returns, and other non-standard entries) were excluded, meaning this analysis reflects completed sales only, not net sales after returns.
- The two negative-UnitPrice “Adjust bad debt” records were excluded as accounting adjustments rather than sales transactions.
- The Country field is heavily dominated by the United Kingdom, which limits the ability to draw balanced conclusions across countries.
- The relationship observed between quantity sold and revenue is correlational and should not be interpreted as causal.

SECTION 12

Project Workflow

The project was organised into four sequential notebooks:

Notebook	Purpose
01_data_understanding.ipynb	Initial inspection of the raw dataset: structure, missing values, and descriptive statistics.
02_data_cleaning.ipynb	Cleaning of the raw dataset, including handling of negative values, duplicates, and missing identifiers; export of the cleaned dataset.
03_exploratory_analysis.ipynb	Derivation of the Revenue field and computation of monthly revenue, monthly quantity, and top revenue-generating products.
04_matplotlib_visualization.ipynb	Creation of Matplotlib visualizations summarising the exploratory analysis results.

SECTION 13

Tools & Technologies

Tool	Role in Project
Python	Core programming language used for data cleaning and analysis
Pandas	Data manipulation, cleaning, and aggregation
Matplotlib	Data visualization
Jupyter Notebook	Interactive development and documentation environment
Excel (.xlsx)	Format used for exporting the cleaned dataset

SECTION 14

Conclusion

This project applied a structured data cleaning and exploratory analysis process to a 541,909-row online retail transaction dataset, resulting in a cleaned dataset of 401,604 records suitable for sales analysis. The analysis quantified total revenue of £8,278,519.42 over the observed period, identified November 2011 as the strongest complete month and April 2011 as the weakest, and surfaced the top 10 revenue-generating products, led by REGENCY CAKESTAND 3 TIER.

Throughout the analysis, data quality issues — including missing identifiers, negative Quantity and UnitPrice values, and duplicate records — were identified and handled transparently, with each decision documented against the underlying data. All findings in this report are drawn directly from the computations performed in the accompanying notebooks.

SECTION 15

Future Scope

The current analysis is descriptive and focuses on revenue and quantity trends at the monthly and product level. Potential directions for extending this work include:

- Customer-level analysis (e.g., purchase frequency or customer segmentation) using the CustomerID field retained in the cleaned dataset.
- Incorporating cost data, where available, to move from revenue analysis to profitability analysis.
- Further investigation into the returns/cancellation records (InvoiceNo beginning with “C”) that were excluded from this analysis.
- Extending the analysis to a full December period once complete data becomes available, to allow direct comparison with other calendar months.
- Building an interactive dashboard to make the monthly and product-level findings more accessible to non-technical stakeholders.

— End of Report —